Advanced Statistical Methods II

Lecture VI: EM for Incomplete Tables and General Missing-Data Likelihoods

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Outline of Today's Lecture

Where we are after Lecture V

In Lecture V, we applied EM to finite normal mixtures.

- Observed data: numerical observations $Y = (Y_1, \dots, Y_n)$.
- Missing data: latent component labels $Z = (Z_1, \dots, Z_n)$.
- E-step: compute responsibilities

$$\tau_{ik}^{(t)} = \mathbb{P}_{\theta^{(t)}}(Z_i = k \mid Y_i).$$

- M-step: replace missing indicators by responsibilities and use complete-data MLE formulas.

missing complete-data sufficient statistics are replaced by conditional expectations.

Today's lecture

Today we move from hidden labels in mixtures to incomplete categorical data and general missing-data likelihoods.

> If the observed table is only a partial view of the full table, can we still do likelihood-based estimation as if the table were complete?

What makes a table incomplete?

A categorical table becomes incomplete when we do not know the exact category for every unit.

Examples

- In a contingency table, some records have row variable observed but column variable missing.
- In a classification problem, we may know that an observation belongs to a subset of categories, but not the exact category.

For multinomial models, the complete-data sufficient statistics are cell counts. If some cell memberships are unknown, EM estimates the missing parts of those counts.

Complete multinomial counts

Suppose each unit belongs to one of K categories.

$$\mathbb{P}(\text{category } k) = p_k, \qquad p_k > 0, \qquad \sum_{k=1}^{K} p_k = 1.$$

If the full cell counts are observed,

$$N_k = \#\{\text{units in category } k\}, \qquad \sum_{k=1}^{K} N_k = N,$$

then

$$(N_1, \ldots, N_K) \sim \mathrm{Multinomial}(N; p_1, \ldots, p_K).$$

Ignoring constants, the complete-data log-likelihood is

$$\ell_c(\boldsymbol{p}) = \sum_{k=1}^{K} N_k \log p_k.$$

See, e.g., Agresti (2013), Ch. 1.

Complete-data MLE for the multinomial model

Maximize

$$\sum_{k=1}^{K} N_k \log p_k \quad \text{s.t} \quad \sum_{k=1}^{K} p_k = 1.$$

Set

$$\mathcal{L}(\boldsymbol{p}, \lambda) = \sum_{k=1}^{K} N_k \log p_k + \lambda \left(1 - \sum_{k=1}^{K} p_k \right).$$

Then

$$\frac{\partial \mathcal{L}}{\partial p_k} = \frac{N_k}{p_k} - \lambda = 0 \quad \Rightarrow \quad p_k = \frac{N_k}{\lambda}.$$

Using $\sum_k p_k = 1$, we get $\lambda = N$, hence $\widehat{p}_k = N_k / N$.

A simple incomplete table

Consider three categories with probabilities

$$\boldsymbol{p} = (p_1, p_2, p_3), \qquad p_1 + p_2 + p_3 = 1.$$

Observed information:

Observed information	**Meaning**	**Count**
Category 1 exactly	definitely in cell 1	n_1
Category 2 exactly	definitely in cell 2	n_2
Category 3 exactly	definitely in cell 3	n_3
Category 1 or 2	exact cell unknown	m

Total sample size:

$$N = n_1 + n_2 + n_3 + m.$$

The m ambiguous observations are known to be in $\{1, 2\}$, but their exact split is missing.

Observed-data likelihood for the simple table

For an observation known exactly to be in category k, the probability contribution is p_k.

For an observation known only to be in category 1 or 2, the probability contribution is

$$p_1 + p_2.$$

Therefore, ignoring multinomial constants, the observed-data likelihood is

$$L(\boldsymbol{p}; Y) \propto p_1^{n_1} p_2^{n_2} p_3^{n_3} (p_1 + p_2)^m.$$

The observed log-likelihood is

$$\ell(\boldsymbol{p}; Y) = n_1 \log p_1 + n_2 \log p_2 + n_3 \log p_3 + m \log(p_1 + p_2).$$

Complete-data formulation

Let

U = number of ambiguous observations that truly belong to category 1.

Then

$m - U$ = number of ambiguous observations that truly belong to category 2.

If U were known, the complete cell counts would be

$$N_1 = n_1 + U, \qquad N_2 = n_2 + (m - U), \qquad N_3 = n_3.$$

The complete-data log-likelihood would be

$$\ell_c(\boldsymbol{p}; Y, U) = (n_1 + U) \log p_1 + (n_2 + m - U) \log p_2 + n_3 \log p_3.$$

The complete-data MLE would simply be complete cell count divided by N.

Distribution of the missing split

Given that an ambiguous observation lies in $\{1, 2\}$, its conditional probability of being in category 1 is

$$\mathbb{P}_{\boldsymbol{p}}(1 \mid 1 \text{ or } 2) = \frac{p_1}{p_1 + p_2}.$$

Therefore, at iteration t,

$$U \mid Y, \boldsymbol{p}^{(t)} \sim \text{Binomial}\left(m, \frac{p_1^{(t)}}{p_1^{(t)} + p_2^{(t)}}\right).$$

Hence the E-step gives

$$\widehat{U}^{(t)} = \mathbb{E}_{\boldsymbol{p}^{(t)}}(U \mid Y) = m\frac{p_1^{(t)}}{p_1^{(t)} + p_2^{(t)}}.$$

Similarly,

$$\mathbb{E}_{\boldsymbol{p}^{(t)}}(m - U \mid Y) = m\frac{p_2^{(t)}}{p_1^{(t)} + p_2^{(t)}}.$$

E-step for the simple table

The expected complete counts at iteration t are:

$$\widetilde{N}_1^{(t)} = n_1 + m\frac{p_1^{(t)}}{p_1^{(t)} + p_2^{(t)}},$$

$$\widetilde{N}_2^{(t)} = n_2 + m\frac{p_2^{(t)}}{p_1^{(t)} + p_2^{(t)}},$$

$$\widetilde{N}_3^{(t)} = n_3.$$

Thus

$$Q(\boldsymbol{p} \mid \boldsymbol{p}^{(t)}) = \sum_{k=1}^{3} \widetilde{N}_k^{(t)} \log p_k.$$

This is exactly the complete multinomial log-likelihood, with cell counts replaced by expected cell counts.

M-step for the simple table

The M-step maximizes

$$Q(\boldsymbol{p} \mid \boldsymbol{p}^{(t)}) = \sum_{k=1}^{3} \widetilde{N}_k^{(t)} \log p_k$$

subject to $p_1 + p_2 + p_3 = 1$.

Therefore, using the same multinomial MLE argument,

$$p_1^{(t+1)} = \frac{\widetilde{N}_1^{(t)}}{N},$$

$$p_2^{(t+1)} = \frac{\widetilde{N}_2^{(t)}}{N},$$

$$p_3^{(t+1)} = \frac{\widetilde{N}_3^{(t)}}{N} = \frac{n_3}{N}.$$

The M-step treats the expected completed table as if it were the completed table.

Numerical example: data and initialization

Suppose

$$n_1 = 18, \qquad n_2 = 12, \qquad n_3 = 20, \qquad m = 10.$$

Then

$$N = 18 + 12 + 20 + 10 = 60.$$

Start from

$$\boldsymbol{p}^{(0)} = (1/3, 1/3, 1/3).$$

The observed log-likelihood, ignoring constants, is

$$\ell(\boldsymbol{p}) = 18 \log p_1 + 12 \log p_2 + 20 \log p_3 + 10 \log(p_1 + p_2).$$

How does EM split the 10 ambiguous observations between categories 1 and 2?

Numerical example: first EM iteration

At $\boldsymbol{p}^{(0)} = (1/3, 1/3, 1/3)$,

$$\widehat{U}^{(0)} = 10\frac{1/3}{1/3 + 1/3} = 5.$$

So the expected complete counts are

$$\widetilde{N}_1^{(0)} = 18 + 5 = 23,$$

$$\widetilde{N}_2^{(0)} = 12 + 5 = 17,$$

$$\widetilde{N}_3^{(0)} = 20.$$

The M-step gives

$$\boldsymbol{p}^{(1)} = \frac{1}{60}(23, 17, 20) = (0.3833, 0.2833, 0.3333).$$

Numerical example: EM trace

Iteration	$\widehat{U}^{(t)}$	$p_1^{(t+1)}$	$p_2^{(t+1)}$	$p_3^{(t+1)}$	$\ell(\boldsymbol{p}^{(t+1)})$
0	5.0000	0.3833	0.2833	0.3333	-64.420
1	5.7500	0.3958	0.2708	0.3333	-64.380
2	5.9375	0.3990	0.2677	0.3333	-64.377
3	5.9844	0.3997	0.2669	0.3333	-64.377

EM monotonically increases the observed-data log-likelihood and approaches $(0.4000, 0.2667, 0.3333)$.

Likelihood values ignore multinomial constants.

Why the limiting answer is intuitive

The observed data tell us two things.

1. Category 3 is exactly observed:
$$\widehat{p}_3 = \frac{20}{60} = \frac{1}{3}.$$
2. Among observations known to be in categories 1 or 2, the exact observed split is
$$\frac{n_1}{n_1 + n_2} = \frac{18}{30} = 0.6.$$

Since
$$\widehat{p}_1 + \widehat{p}_2 = 1 - \widehat{p}_3 = \frac{40}{60},$$
we get
$$\widehat{p}_1 = 0.6 \times \frac{40}{60} = 0.4, \qquad \widehat{p}_2 = 0.4 \times \frac{40}{60} = 0.2667.$$

The EM algorithm reaches the same maximum likelihood answer through repeated expected completions.

A Problem: overlapping ambiguous categories

Suppose there are three categories with probabilities

$$\boldsymbol{p} = (p_1, p_2, p_3), \qquad p_1 + p_2 + p_3 = 1.$$

The observed data are:

Observed information	**Count**
exactly category 1	n_1
exactly category 2	n_2
exactly category 3	n_3
category 1 or category 2, but not known which	m_{12}
category 2 or category 3, but not known which	m_{23}

Derive one EM update for (p_1, p_2, p_3).

Solution to the problem

At iteration t, split each ambiguous group conditionally using the current probabilities.

$$\widetilde{u}_{12}^{(t)} = \mathbb{E}_{\boldsymbol{p}^{(t)}}(\text{number from } m_{12} \text{ in category } 1 \mid Y) = m_{12} \frac{p_1^{(t)}}{p_1^{(t)} + p_2^{(t)}}.$$

Hence the expected number from m_{12} in category 2 is

$$m_{12} - \widetilde{u}_{12}^{(t)} = m_{12} \frac{p_2^{(t)}}{p_1^{(t)} + p_2^{(t)}}.$$

Similarly,

$$\widetilde{u}_{23}^{(t)} = m_{23} \frac{p_2^{(t)}}{p_2^{(t)} + p_3^{(t)}}$$

for the expected number from m_{23} in category 2, and

$$m_{23} - \widetilde{u}_{23}^{(t)} = m_{23} \frac{p_3^{(t)}}{p_2^{(t)} + p_3^{(t)}}$$

for category 3.

The expected complete counts are

$$\widetilde{N}_1^{(t)} = n_1 + \widetilde{u}_{12}^{(t)},$$

$$\widetilde{N}_2^{(t)} = n_2 + \{m_{12} - \widetilde{u}_{12}^{(t)}\} + \widetilde{u}_{23}^{(t)},$$

$$\widetilde{N}_3^{(t)} = n_3 + \{m_{23} - \widetilde{u}_{23}^{(t)}\}.$$

Let

$$N = n_1 + n_2 + n_3 + m_{12} + m_{23}.$$

Then the M-step is just normalization:

$$p_k^{(t+1)} = \frac{\widetilde{N}_k^{(t)}}{N}, \qquad k = 1, 2, 3.$$

The trick is that category 2 receives expected contributions from *two different ambiguous groups*.

Now suppose each unit has two categorical variables.

For example:

	Test +	**Test −**
Disease +	n_{11}	n_{12}
Disease −	n_{21}	n_{22}

Let

$$p_{ij} = \mathbb{P}(\text{row } i, \text{ column } j), \qquad \sum_{i,j} p_{ij} = 1.$$

With complete counts,

$$\ell_c(\boldsymbol{p}) = \sum_{i,j} n_{ij} \log p_{ij}, \qquad \widehat{p}_{ij} = \frac{n_{ij}}{N}.$$

Different kinds of incomplete records

For a two-way table, a record may be incomplete in several ways.

Record type	What is observed?
Complete	row and column are both observed
Row-only	row observed, column missing
Column-only	column observed, row missing
Neither observed	both row and column missing

Let

- c_{ij}: fully observed count in cell (i, j),
- a_i: row i observed but column missing,
- b_j: column j observed but row missing,
- r: neither row nor column observed.

Each incomplete record contributes to several possible cells. EM assigns fractional expected counts to those cells.

E-step: row-only records

Consider the a_i records for which row i is known, but column is missing.

Given row i, the conditional probability of column j is

$$\mathbb{P}_{\boldsymbol{p}}(\text{column } j \mid \text{row } i) = \frac{p_{ij}}{p_{i+}},$$

where

$$p_{i+} = \sum_j p_{ij}.$$

Therefore, at iteration t, the expected contribution to cell (i, j) from row-only records is

$$a_i \frac{p_{ij}^{(t)}}{p_{i+}^{(t)}}.$$

Row-only records are distributed across the row according to the current conditional column probabilities.

E-step: column-only and fully missing records

For the b_j records with column j observed but row missing,

$$\mathbb{P}_{\boldsymbol{p}}(\text{row } i \mid \text{column } j) = \frac{p_{ij}}{p_{+j}},$$

where

$$p_{+j} = \sum_i p_{ij}.$$

So their expected contribution to cell (i, j) is

$$b_j \frac{p_{ij}^{(t)}}{p_{+j}^{(t)}}.$$

For the r records with neither row nor column observed, the expected contribution to cell (i, j) is simply

$$r p_{ij}^{(t)}.$$

Fully missing records add very little information about the relative cell probabilities; they mostly affect total sample size.

Expected completed table

Putting the pieces together, the expected complete count in cell (i, j) is

$$\widetilde{n}_{ij}^{(t)} = c_{ij} + a_i \frac{p_{ij}^{(t)}}{p_{i+}^{(t)}} + b_j \frac{p_{ij}^{(t)}}{p_{+j}^{(t)}} + r p_{ij}^{(t)}.$$

Then

$$Q(\boldsymbol{p} \mid \boldsymbol{p}^{(t)}) = \sum_{i,j} \widetilde{n}_{ij}^{(t)} \log p_{ij}.$$

The M-step is therefore

$$p_{ij}^{(t+1)} = \frac{\widetilde{n}_{ij}^{(t)}}{N}.$$

The table version of EM is especially transparent: incomplete records are redistributed into cells using current conditional probabilities.

A small two-way numerical setup

Suppose the complete observed records are

	Column 1	**Column 2**
Row 1	12	8
Row 2	6	14

Additionally:

- $a_1 = 5$: row 1 observed, column missing;
- $a_2 = 3$: row 2 observed, column missing;
- $b_1 = 4$: column 1 observed, row missing;
- $b_2 = 2$: column 2 observed, row missing;
- $r = 0$: no records with both variables missing.

Total sample size:

$$N = 12 + 8 + 6 + 14 + 5 + 3 + 4 + 2 = 54.$$

One E-step from proportional starting values

Use the complete observed table as a starting point:

$$\boldsymbol{p}^{(0)} = \frac{1}{40}\begin{pmatrix} 12 & 8 \\ 6 & 14 \end{pmatrix} = \begin{pmatrix} 0.30 & 0.20 \\ 0.15 & 0.35 \end{pmatrix}.$$

Then

$$p_{1+}^{(0)} = 0.50, \qquad p_{2+}^{(0)} = 0.50,$$
$$p_{+1}^{(0)} = 0.45, \qquad p_{+2}^{(0)} = 0.55.$$

For example, the row-1 missing-column records are split as

$$5 \times \frac{0.30}{0.50} = 3.0 \quad \text{and} \quad 5 \times \frac{0.20}{0.50} = 2.0.$$

The same idea is applied to every incomplete group of records.

Expected counts in the small two-way example

Using the formula

$$\widetilde{n}_{ij}^{(0)} = c_{ij} + a_i \frac{p_{ij}^{(0)}}{p_{i+}^{(0)}} + b_j \frac{p_{ij}^{(0)}}{p_{+j}^{(0)}},$$

we get approximately

	Column 1	**Column 2**
Row 1	$12 + 3.00 + 2.67 = 17.67$	$8 + 2.00 + 0.73 = 10.73$
Row 2	$6 + 0.90 + 1.33 = 8.23$	$14 + 2.10 + 1.27 = 17.37$

The M-step gives

$$p_{ij}^{(1)} = \frac{\widetilde{n}_{ij}^{(0)}}{54}.$$

Thus

$$\boldsymbol{p}^{(1)} \approx \begin{pmatrix} 0.3272 & 0.1988 \\ 0.1525 & 0.3217 \end{pmatrix}.$$

Problem 2: mixed missingness in a 2×2 table

Let the complete cell probabilities be p_{ij}, $i, j = 1, 2$, with

$$\sum_{i=1}^{2} \sum_{j=1}^{2} p_{ij} = 1.$$

Observed data contain:

- fully observed counts n_{ij};
- row-only records: r_i, row i observed but column missing;
- column-only records: c_j, column j observed but row missing;
- fully missing records: f, neither row nor column observed.

Write the E-step expected complete count $\widetilde{n}_{ij}^{(t)}$, and then the M-step.

Solution to Problem 2: general formula

At iteration t, define the current row and column margins

$$p_{i+}^{(t)} = p_{i1}^{(t)} + p_{i2}^{(t)}, \qquad p_{+j}^{(t)} = p_{1j}^{(t)} + p_{2j}^{(t)}.$$

Then

$$\widetilde{n}_{ij}^{(t)} = n_{ij} + r_i \frac{p_{ij}^{(t)}}{p_{i+}^{(t)}} + c_j \frac{p_{ij}^{(t)}}{p_{+j}^{(t)}} + f p_{ij}^{(t)}.$$

The four terms have clear meanings:

- observed complete count;
- expected allocation of row-only records across columns;
- expected allocation of column-only records across rows;
- expected allocation of fully missing records across all cells.

EM for exponential-family complete data

Many complete-data models can be written as

$$\ell_c(\theta; Y, Z) = \eta(\theta)^T S(Y, Z) - A(\theta) + C(Y, Z),$$

where

- $S(Y, Z)$ is the complete-data sufficient statistic,
- $\eta(\theta)$ is the natural parameter,
- $A(\theta)$ is the log-partition function.

Then the EM Q-function is

$$Q(\theta \mid \theta^{(t)}) = \eta(\theta)^T \mathbb{E}_{\theta^{(t)}}\{S(Y, Z) \mid Y\} - A(\theta) + \text{constant}.$$

In exponential-family problems, the E-step often reduces to computing the expected complete-data sufficient statistic.

1. **Choose the complete data.**

$$X = (Y, Z).$$

2. **Write the complete-data log-likelihood.**

$$\ell_c(\theta; Y, Z).$$

3. **Identify complete-data sufficient statistics.**

$$S(Y, Z).$$

4. **Compute their conditional expectations.**

$$\widetilde{S}^{(t)} = \mathbb{E}_{\theta^{(t)}}\{S(Y, Z) \mid Y\}.$$

5. **Maximize as if $\widetilde{S}^{(t)}$ were observed.**

$$\theta^{(t+1)} = \arg\max_{\theta} Q(\theta \mid \theta^{(t)}).$$

The same structure across examples

Problem	Missing part	E-step computes
Poisson hidden source	source labels	expected labels
Normal mixture	component labels	responsibilities
Incomplete multinomial	missing cell split	expected cell counts
Incomplete contingency table	missing row/column	expected cell counts
Bivariate normal with missing values	missing coordinates	conditional moments

The examples look different, but the mathematical pattern is the same: conditional expectation creates a surrogate completed dataset.

EM for incomplete tables: algorithm

Input

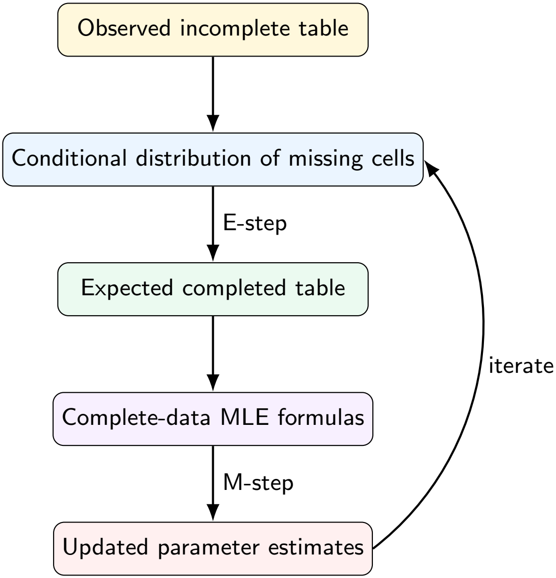
Observed incomplete table and an initial probability table $\boldsymbol{p}^{(0)}$.

For $t = 0, 1, 2, \ldots$:

1. **E-step:** compute expected complete cell counts
$$\widetilde{n}_{ij}^{(t)} = \mathbb{E}_{\boldsymbol{p}^{(t)}}(n_{ij}^{\text{complete}} \mid Y).$$
2. **M-step:** update probabilities
$$p_{ij}^{(t+1)} = \frac{\widetilde{n}_{ij}^{(t)}}{N}.$$
3. **Check convergence:** stop when parameter changes or observed likelihood changes are small.

The whole algorithm is "fill expected counts, then normalize".

Summary


Observed incomplete table
Conditional distribution of missing cells
E-step
Expected completed table
Complete-data MLE formulas
M-step
Updated parameter estimates
iterate

Key takeaway messages

1. In incomplete tables, the complete-data sufficient statistics are usually cell counts.
2. The E-step computes expected missing cell counts using the current parameter estimates.
3. The M-step uses the ordinary complete-data MLE formulas with expected counts.
4. The same principle extends to general missing-data likelihoods through expected sufficient statistics.
5. EM gives maximum likelihood estimates, but standard errors require additional observed-information calculations or resampling.
6. EM is principled only relative to the assumed missing-data model and missingness mechanism.

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